Editor's Suggestion

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Characterizing a Lloyd's mirror interferometer

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Lloyd's mirror interferometers are optical devices used in and relevant to many fields of research today, such as underwater acoustics, sonar research, and laser research. Our work sought to construct a Lloyd's mirror to justify the device's mechanisms and characterize it. We assembled our Lloyd's mirror by placing a transmitter at a distance $d = 1.0287$ m across from a receiver, with a metallic reflector placed perpendicular at a distance h to induce interference. To characterize our device, we derived a value for the wavelength of the signal emitted by the transmitter by varying h across a set of values 0.117 m to 0.320 m, where we subsequently measured the intensity I of the signal to be between 0.72 mA and 9.40 mA. This allowed us to analyze the interference to compute a value of $\lambda = 3.1 \pm 0.6$ cm for the wavelength.

I. INTRODUCTION

Despite having been conducted almost two hundred years ago, the Lloyd's mirror optics experiment, conducted by Humphrey Lloyd in 1834, still finds relevance in the modern age [1]. Particularly, through the construction of the interferometer device specifically used in the experiment. These are now known as Lloyd's mirror interferometers, which are still used in or relevant to many contemporary fields of research, such as underwater acoustics and sonar research as well as laser research [2, 3].

As such, our work seeks to construct a simple Lloyd's mirror interferometer and characterize the properties of this interferometer by deriving the wavelength. We will construct our apparatus by having a transmitter, which emits the electromagnetic signal, placed at a distance d directly across from a receiver, which will measure the properties of the electromagnetic signal. We will position a metallic reflector at a distance h from the transmitter and receiver, in order to reflect part of the electromagnetic signal to induce a phase shift relative to the original signal. This will allow us to measure the wavelength of the signal, by varying the distance h in order to characterize the nature of the interference. We will measure this interference by measuring the changes in intensity of the received electromagnetic signal.

Given this, in this paper we will cover some of the basics of electromagnetic waves and interferometry, to give context to some of the physical and theoretical motivations behind our experiment. We will also cover the nature of Lloyd's mirror interferometers in more depth to shed light on how they function, and to give more context to the reasoning of our experiment. Then, we will cover the methodology of our experiment and its results. We conclude with ideas for further expanding the results of this research, such as confirming results with alternative methods of conduct and measurement.

II. BACKGROUND

When electromagnetic waves reflect off of flat, metallic surfaces, the wave will reflect at an angle that is equal to the angle of the wave at incidence, relative to the normal of the metal surface. That is to say, we may state eq. (1), with θ_r is the angle of reflection and θ_i is the angle of incidence

$$\theta_r = \theta_i. \tag{1}$$

Now let's presume we have a spread of electromagnetic waves, each of which is propagating out from a source. If some of these waves reflect off of a metallic surface, they will become phase shifted relative to the original propagating waves and eventually rejoin them. When there are multiple electromagnetic waves in the same spacial region, they will combine to produce an oscillating electromagnetic field that is equal to the sum of all the individual waves. This leads to two types of interference when electromagnetic fields of differing wavelength combine: *constructive interference* where the waves combine maximally to increase their magnitude and *destructive interference* where the waves cancel maximally such that their magnitude is minimized.

With the appropriate measuring techniques, we may use the nature of constructive and destructive interference in order to make determinations about the nature and properties of the originating electromagnetic wave. This is known as *interferometry.* In our experiment, we have utilized a simple interferometry device known as Lloyd's mirror, a schematic of which is provided in fig. 1.

The Lloyd's mirror functions by having a transmitter, which emits the electromagnetic signal, placed at a distance d directly across from a receiver, which measures properties of the electromagnetic signal. Placed a distance h from the transmitter and receiver is a reflector,

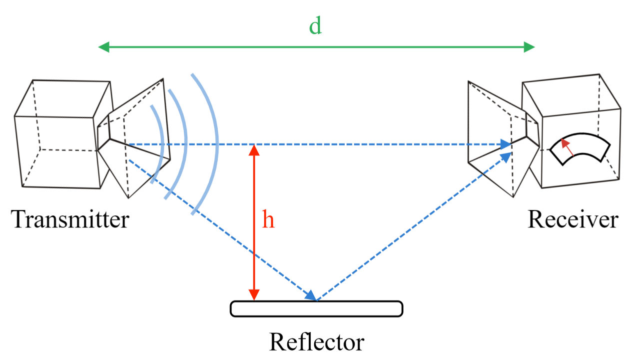

FIG. 1. A diagram of the Lloyd's mirror interferometer setup used within our experiment. A transmitter emits an electromagnetic wave field, where one component of the signal transmits directly to a receiver and another component reflects off of a metallic reflector. This causes interference that changes based on the varying length h. The transmitter is placed in line with the receiver such that there is a fixed distance d between the point of emission and the point of receival in the receiver.

which reflects part of the electromagnetic signal such that it interferes with the incident signal being picked up by the receiver. Due to the difference in path length between the two signals, varying the distance h adjusts the nature of the interference. Specifically, by presuming the signal reflects at the half-way point between the transmitter and receiver and utilizing Pythagorean theorem, we may describe the relation by eq. (2), where Δx denotes the path difference

$$\Delta x = 2\sqrt{h^2 + \left(\frac{d}{2}\right)^2} - d. \tag{2}$$

If Δx is equal to a value $n\lambda$, where n is an integer and λ denotes the wavelength of the incident wave, the interference will be constructive. Similarly, if Δx is equal to a value $\frac{n}{2}\lambda$ the interference will be destructive.

Therefore, we may characterize the transmitter by measuring the wavelength of the signal via studying the interference as h is varied. There are a multitude of ways to accomplish this, but for our experiment the simplest method is via measuring the intensity of the wave measured by the receiver.

Recall that the magnitude of the electric and magnetic field components of an electromagnetic wave are related to the intensity of the wave via the following equation, where I denotes the intensity, c denotes the speed of light, B denotes the magnetic field, μ_0 denotes vacuum permeability, and E denotes the electric field

$$I = \frac{cB^2}{2\mu_0} = \frac{E^2}{2\mu_0 c}. \tag{3}$$

Given intensity's strong relationship to the magnitude of E and B and thus to the amplitude of the wave, points of interference will be easily measured through peaks and dips in the intensity in relation to h. As such, for our investigation, we have chosen to vary h while measuring the intensity measured by the receiver from the transmitter, following the procedure described in the following section.

III. EXPERIMENTAL FINDINGS

A. Methodology

For our experiment, we chose to arrange our setup according to the manner shown in fig. 1, with a two meter ruler affixed to a flat surface in order to line the transmitter and receiver and a rail mechanism lined parallel with the reflector attached, in order to make the varying of the length h more consistent. We prepared the interferometer with a distance d of 1.0287 m. For our measuring procedure, multiple measurements of the intensity for each length of h were taken by differing team members at differing times, and the true intensity was taken to be the average value between them. In the final experiment, we varied h across a set of values 0.117 m to 0.320 m, where we subsequently measured the intensity I to be between 0.72 mA and 9.40 mA. This gave the final results shown in fig. 2.

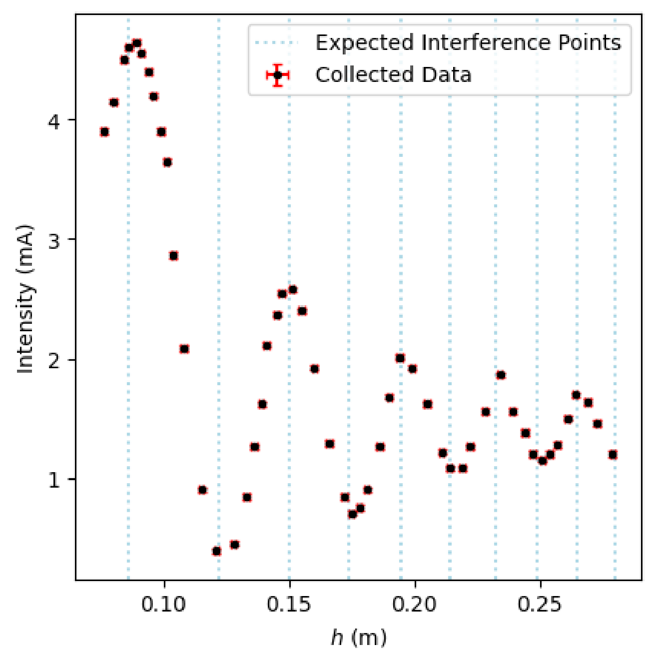

FIG. 2. Graph of intensity in milliamps plotted against h in meters. The points plotted in black indicate the collected data, with respective error bars in red. The dotted lines in a lighter blue color indicate the expected points of constructive and destructive interference, based on eq. (2).

B. Wavelength Results

To find an estimate for our wavelength, we took our points of data at the peaks and troughs seen in fig. 2 and from them calculated the wavelength using eq. (2). We took the true wavelength to be the average of these wavelengths. As such, we derived a result of

$$\lambda = 0.031 \pm 0.006 \text{ m}. \tag{4}$$

Our transmitter was classified as having a wavelength of 0.029 m, according to its specification. As such, our derived result is in relatively close agreement with the expected value.

The small deviation we see in our result may be relatively easily explained by our previous assumption that the interfering waves are being reflected at a point perfectly situated at $d/2$, which is not necessarily true. The majority of the electromagnetic waves are being reflected at this point as such most deviations are negligible, but there are still some waves which are reflected at other points on the reflector. Additionally, it must also be considered that our data may have been insufficient to fully describe each of the peaks and troughs caused by interference, meaning collection of additional data points may help to better describe the electromagnetic signal and the wavelength.

Therefore, further analysis could seek to account for all of the potential points of reflection on the reflector, rather than just the point $d/2$. It may also be beneficial to investigate alternative methods of measurement, such as utilizing a mirror system (as seen in [2]) in order to more clearly map the fringes of the interference. Such examinations would allow to us to more firmly establish our results.

IV. CONCLUSION

In conclusion, our work sought to construct a Lloyd's mirror interferometer and characterize this device by measuring the wavelength of the transmitter. We constructed an interferometer which functions by having a transmitter placed at a distance d directly across from a receiver, with a metallic reflector placed perpendicular at a distance h in order to induce interference. We arranged our setup with a two meter ruler affixed to a flat surface in order to line the transmitter and receiver and measure the distance d, which ended up being 1.0287 m. We additionally employed a rail mechanism lined parallel with the reflector attached, in order to make the varying of the length h more consistent. For our measuring procedure, we made multiple measurements of the intensity for each length of h presumed the true length to be the average of these values. In the final experiment, we varied h across a set of values 11.7 cm to 32.0 cm, where we subsequently measured the intensity I to be between 0.72 mA and 9.40 mA. These experimental results granted a value of $\lambda = 3.1 \pm .6$ cm for the wavelength. This was in close approximation with the expected result of 2.9 cm.

Given our results, we have determined that further analysis could be conducted to better characterize and account for more factors affecting the interference of the signal. In particular, it may be beneficial to consider the effects of the signal reflecting on the metallic reflector at all potential points, rather than just the point $d/2$. It may also be worthwhile to investigate alternative methods of measurement and experimentation in order to confirm the results obtained in this study. Such alternatives could potentially consist of utilizing a mirror system (as seen in [2]) in order to more clearly map the fringes of the interference.

Appendix A: Uncertainty Estimations

a. Distance d uncertainty Both the transmitter and receiver, rather than outputting and receiving signal at the exact emission and receival points, may be treated as being point emitters and receivers and a point 5 cm in from the entry point of the cone, as depicted in fig. 1. As such, we have used this value to describe the potential uncertainty in our derived value of d.

b. Distance h uncertainty In our procedure, we were careful to consider the effects of parameters such as the transmitter/receiver horn width and the reflector width in our estimation of h. As such, we will able to constrain all h measurements to a uniform uncertainty of 0.001 m.

c. Intensity Uncertainty The equipment we used to measure uncertainty with the receiver was highly sensitive and we took care to take multiple measurements of the intensity for each potential data point. As such, we were able to constrain our uncertainties in the average intensity to a value of 0.02 mA.

Appendix B: Error Calculations

a. Wavelength Error Given the eq. (2), we characterized the uncertainty in our wavelength calculations via eq. (B1), where σ_λ denotes the uncertainty in wavelength, σ_d denotes in the uncertainty in d, and σ_h denotes the uncertainty in h

$$\sigma_\lambda = \sqrt{\frac{4h^2\sigma_d^2}{n^2\left(\frac{d^2}{4}+h^2\right)} + \frac{\sigma_h^2\left(\frac{d}{2\sqrt{\frac{d^2}{4}+h^2}}-1\right)^2}{n^2}}. \tag{B1}$$

b. Average Wavelength Error Given a standard calculation of the average by dividing the sum of the values by the population, the uncertainty formula utilized for our results was eq. (B2), where $\sigma_{\bar{\lambda}}$ denotes the uncertainty in the average wavelength and $\sigma_{\lambda,i}$ denotes the uncertainty in each individual wavelength calculation

$$\sigma_{\overline{\lambda}} = \sqrt{\frac{\sum_i \sigma_{\lambda,i}^2}{n^2}}. \tag{B2}$$

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[3] X. Li, Y. Shimizu, S. Ito, W. Gao, and L. Zeng (2012).